Reg. No. | | | | | | | | | | |

[12ME 72]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER-2015

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.

1. a. A person requires 10,12 &12 units of chemicals A, B & C respectively for his garden A liquid product contains 5, 2 & 1 units of A, B & C respectively per jar A dry product contains 1, 2 & 4 units of A, B & C per carton. If the liquid product sells for Rs 3 per jar and the dry product sells for 2 per carton, how many of each should be purchased to minimize the cost & meet the requirements. Formulate & find the optimal solution by using graphical method. **08**

b. Define operation research. Explain the phases of OR study. **06**

2. a. List the general rules for converting any primal into its dual. **06**

b. Solve the given problem by using Big M. **08**

Max $Z=-2x_1 -x_2$

Show That $3x_1+x_2=3$

$4x_1+3x_2\geq 6$

$x_1+2x_2\leq 4$ & $x_1, x_2\geq 0$

3. a. Explain briefly about unbalanced transportation problem and degenerate case in transportation problem. **04**

b. Find the optimal solution by MODI method for the given transportation problem, use NWCR to find the basic feasible solution. **10**

	1	2	3	4	5	Availability
A	4	3	1	2	6	80
B	5	2	3	4	5	60
C	3	5	6	3	2	40
D	2	4	4	5	3	20
Required	60	60	30	40	10	

4. a. Explain the difference between a transportation problem and an assignment problem. Explain situations where an assignment problem can arise. **06**

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b. A company has one surplus truck in each of the cities A,B,C,D & E and one deject truck in each of the cities 1,2,3,4,5 & 6. The distance between the cities in kms is known as given below. Find optimal solution to the assignment of trucks from surplus to shortage so that the distance covered is minimum. **08**

	1	2	3	4	5	6
A	12	10	15	22	18	8
B	10	18	25	15	16	12
C	11	10	3	8	5	9
D	6	14	10	13	13	12
E	8	12	11	7	13	10

5. **a.** List the assumptions made in the sequencing problems. **04**

b. Use graphical method to minimize the time required to process the following jobs with processing time as indicating. Compute the total elapsed time and ideal time in Tabular column. **10**

	Sequence	A	B	C	D	E
Job 1	Time (hrs)	6	8	4	12	4
	Sequence	B	C	A	D	E
Job 2	Time (hrs)	10	8	6	4	12

6. **a.** Define the following: **05**

i. Pure strategy

ii. Payoff matrix

iii. Fair game

iv. Saddle point

b. Solve the following 2 x5 game by using graphical method **09**

	1	2	3	4	5
1	-5	5	0	-1	8
2	8	-4	-1	6	-5

7. **a.** Define the following. **03**

i. Normal time

ii. Crash time

iii. Free float

b. The following table gives the activities in a construction project & other related information. **11**

Activity	T_o	T_m	t_p
1-2	20	30	46
1-3	9	12	21
2-3	3	5	7
2-4	2	3	4
3-4	1	2	3
4-5	12	18	24

8. a. Explain briefly the main characteristics of queuing system. **06**

b. The rate of arrival of customers at a public telephone booth follows Poisson distribution, with an average time of 10 minutes between one customer and the next. The duration of a phone call is assumed to follow exponential distribution with mean time of 3mins. **08**

i. What is the probability that a person arriving at the booth will have to wait?

ii. What is the average length of the non-empty queues that form from time to time?

iii. The Mahanagar Telephone Nigam Ltd. Will install a 2 nd booth when it is convinced that the customers would expect waiting for at least 3 min for their turn to make a call. By how much time should the flow of customers increase in order to justify a second booth?

iv. Estimate the fraction of a day that the phone will be in use.